

Software Architecture

SE6105

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(BSc Honours in Software Engineering)

SE6105 – Course Content

- Topic 01: Basic concepts & principles about software architecture
- Topic 02: Introduction to Software Architectural pattern
- Topic 03: ADL
- Topic 04: 4+1 Architecture
- Topic 05: Practical approaches & methods for Create & Analyse software architecture
- Topic 06: Quality attributes of software architectures
- Topic 07: Examples in architectural design applications & case studies in software architecture (N tier architecture, SOA, Cloud, etc.)

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Multiple dimensions of Software architecture

Real-World Architectures?

Means only “monolith vs microservices”?

- deployment
- communication
- data handling
- scalability
- integration
- organization of code
- infrastructure
- distributed computing
- reliability
- UI composition
- workflow orchestration

A system can simultaneously be:

- Microservices architecture (deployment)
- Event-driven architecture (communication)
- CQRS (data access)
- Layered architecture (code organization)
- Cloud-native architecture (infrastructure)
- Kubernetes-based deployment (operations)

Major Architecture Categories

Category	Focus	Examples
Deployment Architecture	How software is deployed/executed	Monolith, Microservices
Communication Architecture	How components communicate	REST, Event-Driven, gRPC
Structural / Code Organization	Internal code organization	Layered, Hexagonal
Data Architecture	Data ownership and access	CQRS, Event Sourcing
Integration Architecture	Connecting systems	SOA, ESB
Infrastructure Architecture	Runtime environment	Cloud-native, Serverless
Distributed System Architecture	Scaling across machines	Distributed systems
UI Architecture	Frontend composition	MVC, Micro-frontends
Workflow Architecture	Process coordination	Orchestration, Choreography
Reliability Architecture	Fault tolerance	Circuit Breaker, Bulkhead

1. Deployment Architecture

Focuses on:

- how applications are packaged
- deployed
- scaled
- executed

A. Monolithic Architecture

B. Microservices Architecture

C. Modular Monolith

D. Serverless Architecture

2. Communication Architecture

How services/modules exchange information?

A. Request-Response Architecture

Traditional synchronous communication.

B. Event-Driven Architecture (EDA)

Components communicate via events.

C. Message-Oriented Architecture

Uses queues/messages.

D. gRPC Architecture

High-performance binary communication.

3. Structural / Code Organization Architecture

Focuses on:

- internal software structure
- maintainability
- separation of concerns

A. Layered Architecture

Most common enterprise architecture.

B. Clean Architecture

Separates business logic from frameworks.

C. Hexagonal Architecture

Business logic isolated from external systems.

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D. MVC Architecture

Common in web apps.

4. Data Architecture

Focuses on:

- data consistency
- storage patterns
- querying

A. Database-per-Service

Each microservice owns its database.

B. Shared Database

Multiple modules use same DB.

C. CQRS

Command Query Responsibility Segregation.

D. Event Sourcing

Store events instead of current state.

5. Integration Architecture

A. SOA (Service-Oriented Architecture)

- Enterprise services shared organization-wide.

B. ESB (Enterprise Service Bus)

- Central communication hub.

C. API Gateway Architecture

- Single entry point for clients.

6. Infrastructure Architecture

A. Cloud-Native Architecture

Designed specifically for cloud.

B. Containerized Architecture

Uses Docker containers.

C. Kubernetes Architecture

Container orchestration.

7. UI / Frontend Architecture

A. SPA (Single Page Application)

B. Micro-Frontend Architecture

- Frontend split into independent modules.
- Like microservices for UI.

C. SSR Architecture

- Server-side rendering.
- Example: Next.js

Concern	Architecture
Deployment	Microservices
Communication	Event-driven
Infrastructure	Cloud-native
Reliability	Circuit breakers
Data	Distributed databases
UI	SPA
Integration	API Gateway

Simple E-Commerce System

Concern	Possible Architecture
Deployment	Microservices
Communication	REST + Events
Database	Database-per-service
Frontend	React SPA
Infrastructure	Kubernetes
Reliability	Retry + Circuit breaker
Workflow	Choreography

Architectures are NOT mutually exclusive.

If architecture, like designing a city

Concern	Analogy
Deployment	Building arrangement
Communication	Roads & transport
Data	Storage systems
Reliability	Emergency systems
Infrastructure	Power/water supply
Workflow	Traffic coordination

Multiple dimensions of Software architecture

Thank You!